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作业8.

8.50.

(1). $f_x = 4-2x$, $f_y = -4-2y$.

$\therefore x_0=2$, $y_0=-2$.

$\therefore$ 极大值为 $f(2,-2)=8$.

(3). $f_x = -2x(y-x^4)-4x^3(y-x^2)$

$f_y = y-x^4+y-x^2 = 2y-x^2-x^4$.

$\therefore$ 令 $f_{x0}=f_{y0}=0$.

$\therefore$ 驻点为 $(0,0)$, $(\pm\frac{1}{\sqrt{2}}, \frac{3}{8})$, $(\pm1, 1)$

$f_{xx} = -y(2+12x^2)+30x^4$.

$f_{xy} = -(2x+4x^3)$

$f_{yy} = 2$. 记 $H = f_{xx}f_{yy}-f_{xy}^2$

$(0,0)$: $H=0$ 非极值.

$(\pm\frac{1}{\sqrt{2}}, \frac{3}{8})$: $H=1$, $f_{xx}=\frac{9}{2}>0$

$\therefore$ 为极小值点, $f=-\frac{1}{64}$

$(\pm1, 1)$: $H=-4<0$ $\therefore$ 非极值.

$\therefore f(\pm\frac{1}{\sqrt{2}}, \frac{3}{8}) = -\frac{1}{64}$ 为极小值.

57. (1). $f_x = 2x$, $f_y = -2y$.

$f_{xx}=2$, $f_{xy}=0$, $f_{yy}=-2$.

$H=-4<0$.

边界: $x^2+\frac{y^2}{4}=1$ 即 $y^2=4(1-x^2)$.

$g(x)=f=5x^2-2$, $-1\le x\le 1$, $g'(x)=10x$.

$\therefore g(0)=-2$ 为最小值

$g(1)=3$ 为最大值.

52. 圆形内接三角形面积为

$$S=\frac{1}{2}r^2(\sin\theta_1+\sin\theta_2+\sin(2\pi-(\theta_1+\theta_2)))$$

$$=\frac{1}{2}r^2(\sin\theta_1+\sin\theta_2-\sin(\theta_1+\theta_2))$$

$$\frac{\partial S}{\partial\theta_1}=\frac{1}{2}r^2(\cos\theta_1-\cos(\theta_1+\theta_2)),\ \frac{\partial S}{\partial\theta_2}=\frac{1}{2}r^2(\cos\theta_2-\cos(\theta_1+\theta_2)).$$

$\frac{\partial S}{\partial\theta_1}=\frac{\partial S}{\partial\theta_2}=0$ $\therefore \theta_1=\theta_2=\frac{2}{3}\pi$. 此时S取极大值 S_{max}.

对应正三角形 $(\theta_1=\theta_2=2\pi-\theta_1-\theta_2)$.

显然：边界处 $S=0<S_{max}$.

$\therefore$ 得证.

53. 设其为 (x,y).

$$d_1^2=x^2,\ d_2^2=y^2,\ d_3^2=\frac{(2x+y-16)^2}{5}.$$

记 $D=\sum d^2=x^2+y^2+\frac{1}{5}(2x+y-16)^2$.

$$\therefore \frac{\partial D}{\partial x}=\frac{18}{5}x+\frac{4}{5}y-\frac{64}{5},\quad \frac{\partial D}{\partial y}=\frac{4}{5}x+\frac{12}{5}y-\frac{32}{5}.$$

令 $\frac{\partial D}{\partial x}=\frac{\partial D}{\partial y}=0$. $\because \lim_{x,y\to\infty}D=\infty$ $\therefore$ 其必为最小值点.

$$\therefore \begin{cases}9x+2y=32\\ x+3y=8\end{cases} \quad \therefore \begin{cases}x_0=\frac{16}{5}\\ y_0=\frac{8}{5}\end{cases}.$$

即：定为 $(\frac{16}{5},\frac{8}{5})$.

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54.(1). $2x+2y+2xy'+4yy'=0$.

$\therefore \frac{dy}{dx}=-\frac{x+y}{x+2y}$

令 $\frac{dy}{dx}=0 \therefore x=-y$. $\therefore x^2=1$

$\therefore$ 驻点为 $(1,-1)$ 和 $(-1,1)$，经局部验证

$\therefore$ 极大值为 $y=-1$，极小值为 $y=1$

55.(4). $L(x,y,\lambda)=xy+\lambda(x+y-1)$.

$$\begin{cases} L_x=y+\lambda=0 \\ L_y=x+\lambda=0. \\ L_\lambda=x+y-1=0 \end{cases} \quad \therefore \begin{cases} x=\frac{1}{2} \\ y=\frac{1}{2} \end{cases}$$

$\therefore f(\frac{1}{2},\frac{1}{2})=\frac{1}{4}$ 为极大值

(3). $L(x,y,z,\lambda)=x-2y+2z+\lambda(x^2+y^2+z^2-1)$

$$\begin{cases} L_x=1+2\lambda x=0. \\ L_y=-2+2\lambda y=0. \\ L_z=2+2\lambda z=0. \\ L_\lambda=x^2+y^2+z^2-1=0. \end{cases}$$

$$\therefore \begin{cases} x=\frac{1}{3} \\ y=-\frac{2}{3} \\ z=\frac{2}{3} \end{cases} \text{ 或 } \begin{cases} x=-\frac{1}{3} \\ y=\frac{2}{3} \\ z=-\frac{2}{3}. \end{cases}$$

$\therefore f(\frac{1}{3},-\frac{2}{3},\frac{2}{3})=3$ 为极大值

$f(-\frac{1}{3},\frac{2}{3},-\frac{2}{3})=-3$ 为极小值.

56. 椭圆点满足.

$$\begin{cases} z = x^2+y^2. \\ x+y+z-1=0. \Rightarrow z=1-x-y. \end{cases}$$

距离的平方 $D=d^2=x^2+y^2+z^2$

$\therefore D=x^2+y^2+(1-x-y)^2.$

约束条件为 $x^2+y^2+x+y-1=0.$

$L=x^2+y^2+(1-x-y)^2+\lambda(x^2+y^2+x+y-1)$

$$\begin{cases} L_x = 2x-2(1-x-y)+2\lambda x+\lambda=0. \\ L_y=2y-2(1-x-y)+2\lambda y+\lambda=0. \\ L_\lambda = x^2+y^2+x+y-1=0. \end{cases}$$

$\therefore x=y=\dfrac{-1\pm\sqrt{3}}{2}$

$\therefore$ 最大值 $D_{max}=9+5\sqrt{3}$. 在 $x_0=y_0=-\dfrac{1+\sqrt{3}}{2}$ 处取得

最小值 $D_{min}=9-5\sqrt{3}$, 在 $x_0=y_0=\dfrac{-1+\sqrt{3}}{2}$ 处取得

$\therefore d_{max}=\sqrt{9+5\sqrt{3}}$, $d_{min}=\sqrt{9-5\sqrt{3}}$.

57. (1). $D=d^2=(x-1)^2+(y-1)^2+(z-1)^2+(x-2)^2+(y-3)^2+(z-4)^2.$

约束条件 $f(x,z)=3x-2z=0.$

设 $L=D-\lambda f(x,z)$.

$$\begin{cases} L_x=2(x-1)+2(x-2)+3\lambda=0. \\ L_y=2(y-1)+2(y-3)=0. \\ L_z=2(z-1)+2(z-4)-2\lambda=0. \\ L_\lambda=3x-2z=0. \end{cases} \quad \therefore \begin{cases} x=\dfrac{21}{13}. \\ y=2 \\ z=\dfrac{63}{26} \end{cases}$$

$\therefore$ 坐标为 $\left(\dfrac{21}{13}, 2, \dfrac{63}{26}\right)$.

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(3). 设 $L = xyz^3 + \lambda(x^2+y^2+z^2-5R^2)$.

$$\therefore \begin{cases} yz^3+2\lambda x=0. \\ xz^3+2\lambda y=0, \\ 3xyz^2+2\lambda z=0. \\ x^2+y^2+z^2-5R^2=0. \end{cases} \quad x>0,\ y>0,\ z>0.$$

$$\therefore \begin{cases} x=R \\ y=R \\ z=\sqrt{3}R \end{cases}$$ 坐标为 $(R, R, \sqrt{3}R)$.

58. (1). 设抛物线上一点 (x, y).

$$D=d^2=\frac{(x-y-2)^2}{2}.$$

即: 最小化 $F=(x-y-2)^2$; 条件为 $y-x^2=0$.

$\therefore$ 设 $L=(x-y-2)^2+\lambda(y-x^2)$

$$\therefore \begin{cases} 2(x-y-2)-2\lambda x=0. \\ 2(x-y-2)-\lambda=0. \\ y-x^2=0 \end{cases}$$

$$\therefore \begin{cases} x=\frac{1}{2} \\ y=\frac{1}{4} \end{cases}$$ 此时 $d=\frac{7}{8}\sqrt{2}$ 为最小值.

59. $S=ab+2ac+2bc$.

约束: $abc=V$

$\therefore L=(ab+2ac+2bc)+\lambda(abc-V)$

$$\therefore \begin{cases} b+2c+\lambda bc=0. \\ a+2c+\lambda ac=0. \\ 2a+2b+\lambda ab=0 \\ abc-V=0 \end{cases}$$

$$\therefore \begin{cases} a=b=(2V)^{\frac{1}{3}} \\ c=\frac{1}{2}\cdot(2V)^{\frac{1}{3}} \end{cases}$$

其中 a, b 为底面边长, c 为水箱的高

60. $P=x_1p_1+x_2p_2$.

约束: $2x_1^{\alpha}x_2^{1-\alpha}=12$ 即 $x_1^{\alpha}x_2^{1-\alpha}-6=0$.

$L=x_1p_1+x_2p_2+\lambda(x_1^{\alpha}x_2^{1-\alpha}-6)$

$$\therefore \begin{cases} p_1+\lambda x_2^{1-\alpha}\cdot\alpha x_1^{\alpha-1}=0. \\ p_2+\lambda x_1^{\alpha}\cdot(1-\alpha)x_2^{1-\alpha}=0. \\ x_1^{\alpha}x_2^{1-\alpha}-6=0. \end{cases}$$

$$\therefore \begin{cases} x_1=\left(\frac{6p_2\alpha}{p_1(1-\alpha)}\right)^{1-\alpha} \\ x_2=\left(\frac{6p_1(1-\alpha)}{p_2\alpha}\right)^{\alpha}. \end{cases}$$

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思考题

1. 多边形总面积

$$S=\frac{1}{2}\sum_{i=1}^{n}\sin\theta_i$$

约束条件：$\sum_{i=1}^{n}\theta_i-2\pi=0.$

拉格朗日函数：$L=\frac{1}{2}\sum_{i=1}^{n}\sin\theta_i+\lambda\left(\sum_{i=1}^{n}\theta_i-2\pi\right)$

$$\therefore\begin{cases}\frac{\partial L}{\partial\theta_i}=0\\ \frac{\partial L}{\partial\lambda}=0\end{cases}\qquad\therefore\begin{cases}\frac{1}{2}\cos\theta_i+\lambda=0.\\ \sum_{i=1}^{n}\theta_i-2\pi=0.\end{cases}$$

$\therefore\cos\theta_i=-2\lambda$

$\therefore\theta_i$ 均相等，此时 $\theta_i=\frac{2\pi}{n}.$

又$\because$ 边界（$\theta_k\to2\pi$）处 $S\to0.$

$\therefore\theta_1=\theta_2=\cdots=\theta_n=2\pi/n$ 时，P 的面积最大.

即 P 是正多边形

2. p_i 满足归一化条件：$\sum_{i=1}^{n}p_i=1$

$\therefore$ 目标函数 $H=-\sum_{i=1}^{n}p_i\log p_i$

约束：$\sum_{i=1}^{n}p_i-1=0.$

$\therefore L=-\sum_{i=1}^{n}p_i\log p_i+\lambda\left(\sum_{i=1}^{n}p_i-1\right)$

$$\therefore\begin{cases}\frac{\partial L}{\partial p_i}=0\\ \frac{\partial L}{\partial\lambda}=0.\end{cases}\qquad\therefore\begin{cases}-\log p_i-1+\lambda=0 & ①.\\ \sum_{i=1}^{n}p_i-1=0.\end{cases}$$

由①：p_i 均相等 $\therefore p_1=p_2=\cdots=p_n$ 又$\because\sum_{i=1}^{n}p_i=1$

$\therefore p_i=\frac{1}{n}$ $\therefore$ 即为均匀分布.

3. $U(n_1, n_2, n_3) = n_1^{a_1} n_2^{a_2} n_3^{a_3}$

约束条件 $w - p_1 n_1 - p_2 n_2 - p_3 n_3 = 0$.

设 $u = \ln U = a_1 \ln n_1 + a_2 \ln n_2 + a_3 \ln n_3$

$\therefore L = a_1 \ln n_1 + a_2 \ln n_2 + a_3 \ln n_3 + \lambda (w - p_1 n_1 - p_2 n_2 - p_3 n_3)$.

$$\therefore \begin{cases} \frac{a_1}{n_1} - p_1 \lambda = 0. \\ \frac{a_2}{n_2} - p_2 \lambda = 0 \\ \frac{a_3}{n_3} - p_3 \lambda = 0. \\ w - p_1 n_1 - p_2 n_2 - p_3 n_3 = 0. \end{cases}$$

$\therefore \lambda = \frac{1}{w}(a_1 + a_2 + a_3)$.

$\therefore n_1 = \frac{a_1}{\lambda p_1} = \frac{w a_1}{(a_1 + a_2 + a_3) p_1}$

$n_2 = \frac{w a_2}{(a_1 + a_2 + a_3) p_2}$; $n_3 = \frac{w a_3}{(a_1 + a_2 + a_3) p_3}$.

此为效用最大的采购量

此时拉格朗日乘积 λ 表示边际效用 (以 $\ln U$ 计).

即：最优情况下预算增加对函数 $\ln U$ 增加的影响程度.